

Name: _____

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30 minutes maximum. No aids (book, calculator, etc.) are permitted. Show all work and use proper notation for full credit. Answers should be in reasonably-simplified form. 25 points possible.

1. (5 points) Consider the **telescoping** series $\sum_{n=1}^{\infty} \left(\frac{1}{n} - \frac{1}{n+2} \right)$.

(a) Find a simplified formula for S_k , the k th partial sum of the series.

(b) Use your answer from part (a) to determine the sum of the series.

2. (8 points) For each **geometric** series below, determine whether the series converges and **justify** your conclusion. **If** the series converges, determine the sum.

(a) $\sum_{n=1}^{\infty} \frac{3^{2n}}{4^n}$

(b) $\sum_{n=1}^{\infty} \frac{(-2)^{n-1}}{\pi 5^{n-1}}$

3. (8 points) For each series below, determine whether the series converges and **justify** your conclusion. Your justification must include the test or series you are using to draw your conclusion.

(a) $\sum_{n=1}^{\infty} \frac{1}{n\sqrt{6}}$

(b) $\sum_{n=1}^{\infty} \frac{n}{8n+11}$

4. (4 points) Apply the Integral Test to determine if the series $\sum_{n=3}^{\infty} \frac{1}{n\sqrt{\ln(n)}}$ converges or diverges. (You do not have to show that the Integral Test applies; it does.)